

SYNENT Technology Internship

Task 4: Password Generator

1. Introduction

A Python command-line application that generates random passwords using letters, digits, and special characters.

2. Objective

To demonstrate modules, functions, random selection, strings, input validation, and exception handling.

3. Methodology

- Create a character pool from letters, digits, and punctuation.
- Collect and validate the requested password length.
- Select random characters with `random.choice()`.
- Join the characters and display the generated password.

4. Code Review

The original version already used `random.choice()` with lowercase letters, uppercase letters, numbers, and selected special characters. The improved version adds reusable functions, standard string constants, input validation, exception handling, and a clear `main()` entry point.

5. Result

The application successfully generates a random password according to the requested length and handles invalid input safely.

6. Conclusion

The project provides practical experience with Python modules, functions, random selection, strings, and error handling.